MAJID HAJI BAGHERI

Energy Storage · Energy Harvesting & Nanotechnology · MEMS & Nanofabrication · Device Engineering
Cleanroom Process Development · Hardware–ML Integration · Materials R&D

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Work Authorization: *Canadian Open Work permit*

PROFESSIONAL SUMMARY

PhD candidate in Electrical & Computer Engineering specializing in **self-powered sensor systems and energy-harvesting devices**. Designs, fabricates, and characterizes piezoelectric and triboelectric nanogenerators (PENGs/TENGs) and integrates them with machine-learning models for intelligent sensing. **Six years of graduate research experience**, including three years leading industry-collaborative R&D with the **National Research Council of Canada (NRC)**, **Shimco North America, Parthian Battery Novin,** and **Iran Nano Battery Network —** covering self-powered sensors, energy harvesting, and lithium-ion battery development. First author of multiple peer-reviewed publications in *Nano Energy, Energy & Environmental Materials,* and *Advanced Sensor Research*; co-author on a ***Nature Communications*** paper. Strong combined expertise in cleanroom microfabrication, multi-modal characterization, DFT and Monte Carlo simulations, Python-based ML, and translating partner objectives into delivered prototypes.

CORE COMPETENCIES

Energy Harvesting & Sensors: Piezoelectric & triboelectric nanogenerators (PENGs/TENGs); self-powered sensor systems; wearable & cryogenic-compatible devices; lead-free perovskite piezocomposites; lithium-ion battery anodes.

Microfabrication & Cleanroom: Photolithography; sputtering; CVD, PVD, CVT; spin coating; electrospinning; thin-film deposition; etching; bonding; MEMS / diode / MOSFET device fabrication; class-controlled cleanroom operation.

Characterization: XRD; FTIR; SEM/EDX; TEM; AFM; optical microscopy; I–V, C–V; cyclic voltammetry; galvanostatic battery cycling; electromechanical and electroacoustic testing.

Simulation & Modeling: Quantum ESPRESSO (First Principles, DFT, electronic structure, ferroelectric polarization); COMSOL Multiphysics; Ansys Lumerical; Monte Carlo / stochastic growth simulations (KPZ-class thin-film modeling); MATLAB.

Programming, Data & ML: Python (signal processing, ML classification on sensor data); MATLAB; Origin Pro; LaTeX; Office.

Custom Instrumentation: Designed and built motor-tapping energy-harvester test benches, cryogenic / low-temperature testing systems, and oxygen-controlled glove boxes.

Project Leadership: Industry-collaborative project planning; milestone reporting; technical presentations to government and industry partners; mentoring of co-op and undergraduate research assistants.

Certifications: Certified Radiation Protection Professional — Non-Destructive Testing (NDT), Medical and Industrial.

PROFESSIONAL EXPERIENCE

Graduate Researcher (PhD) | *University of Waterloo · Prof. Dayan Ban's Lab* **Sept 2022 – Present**

- **Industry collaboration — National Research Council Canada (NRC, Ottawa):** Led joint development of machine-learning-enabled self-powered sensor systems; delivered scheduled milestone reports, technical presentations, and peer-reviewed publications to NRC partners.
- **Industry collaboration — Shimco North America:** Coordinated joint R&D on energy-harvesting and self-powered sensor technologies; prepared progress reports and delivered technical presentations to industry stakeholders; project resulted in publishable outcomes.
- Designed and fabricated high-performance **piezoelectric and triboelectric nanogenerators (PENGs/TENGs)** for self-powered sensing, wearable electronics, and energy harvesting; results published in *Nano Energy* and *EEM Wiley.*
- Built a **TENG-based acoustic sensor integrated with a machine-learning classifier** for live sound monitoring and captioning, published in *Advanced Sensor Research* and presented at IEEE CCECE.

- Developed **multi-layer cascaded PENG architecture** enabling reliable operation under cryogenic and low-temperature conditions, expanding deployment scenarios for self-powered devices.
- Performed first-principles **DFT simulations (Quantum ESPRESSO)** on perovskite materials to investigate electronic structure, ferroelectric polarization, and structure–property relationships informing material selection.
- Executed full device fabrication workflows: **photolithography, thin-film deposition, electrode patterning, encapsulation, and electromechanical characterization**.
- Designed and built **motor-tapping benches and cryogenic setup** for harsh condition testing of energy-harvesting devices.
- Mentored co-op and Undergraduate Research Assistant (URA) students; managed parallel project workflows across three concurrent collaborator-driven research streams.
- Lab work featured in **CTV News Kitchener** and **University of Waterloo News** following *Nature Communications* publication on perovskite piezocomposites (co-author).

Graduate Researcher (MSc) | *Sharif University of Technology · Battery Research Group* **Feb 2021 – Feb 2022**

- Designed, built, and programmed a **custom in-house glove box** for oxygen- and moisture-sensitive lithium-ion battery fabrication, including sealing, atmosphere control, and instrumentation.
- Synthesized and characterized **SnS_2-based nanostructured anode materials** and freestanding $Ti_3C_2T_x$ MXene / graphene-oxide papers for energy-storage applications.
- Optimized battery performance through binder engineering using PEDOT:PSS, PVDF, and CMC polymer systems.
- Led the battery research group; coordinated experimental planning, project workflows, and student team activities.

EDUCATION

Ph.D., Electrical & Computer Engineering (Nanotechnology) · University of Waterloo, Canada· GPA: 3.70 / 4.00 **Expected NOV-2026**

M.Sc., Physics · Sharif University of Technology (SUT), Iran · GPA: 3.82 / 4.00 **2021**

B.Sc., Physics — Condensed Matter · Iran University of Science & Technology (IUST) · GPA: 3.76 / 4.00 · Ranked #1 in cohort **2018**

SELECTED PUBLICATIONS

Co-author on 16 peer-reviewed publications. Highlights below; full list at Google Scholar.

- A. A. Khan, A. Mathur, L. Yin, M. Almadhoun, J. Yin, **M. Haji Bagheri**, M. F. Al Fattah, N. Yan, B. Zhao, V. Maheshwari, D. Ban, *"Breaking dielectric dilemma via polymer functionalized perovskite piezocomposite with large current density output,"* **Nature Communications**, 15, 9511, 2024. doi.org/10.1038/s41467-024-53846-6
- **M. Haji Bagheri**, A. Rajabi-Abhari, P. Li, A. A. Khan, C. Hao, N. R. Tanguy, D. Ban, L. Yu, N. Yan, *"Self-healable, recyclable, and mechanically robust vitrimer composite for high-performance triboelectric nanogenerators and self-powered wireless electronics,"* **Nano Energy**, 110306, 2024. doi.org/10.1016/j.nanoen.2024.110306
- **M. Haji Bagheri**, A. A. Khan, S. Shahzadi, M. M. Rana, M. S. Hasan, D. Ban, *"Advancements and challenges in molecular/hybrid perovskites for piezoelectric nanogenerator application: A comprehensive review,"* **Nano Energy**, 109101, 2023. doi.org/10.1016/j.nanoen.2023.109101
- **M. Haji Bagheri**, A. Rajabi-Abhari, O. Gibbs, P. Xi, A. A. Khan, F. Huang, Md. S. Hassan, N. Yan, D. Ban, *"Sustainable live sound monitoring and classification system enabled by a triboelectric nanogenerator and machine learning techniques,"* **Energy & Environmental Materials**, e70044, 2025. doi.org/10.1002/eem2.70044

AWARDS & RECOGNITION

- **Best Poster Prize** — Waterloo Nanotechnology Conference (WNC), University of Waterloo, 2024.
- **Waterloo Institute for Nanotechnology (WIN) Nanofellowship Award** · Graduate **Student Research Dissemination (GSRD) Award** · **Faculty of Engineering Award (3× recipient)** — University of Waterloo.
- **Research Publication Award** — University of Waterloo, Awarded by supervisor for outstanding publication.
- **Publication Excellence Award** — University of Waterloo, Prof. Dayan Ban's Lab.
- **Direct Admission to MSc** (SUT, top BSc rank) · Ranked **#1** in BSc Physics cohort, IUST.

INTERESTS

- Amateur astronomy and telescope observation - independent study of stellar phenomena and astronomical imaging
- Piano performance